
SYNTHETIC GLOBAL PRECIPITATION MAPS BY GENERATIVE ADVERSARIAL NETWORKS

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ABSTRACT

We generate global precipitation maps using Generative Adversarial Networks (GANs). The architecture implemented in this study consists of a generator network based on convolutional layers generator and a discriminator network based in deconvolutional layers. Leveraging monthly precipitation data derived from satellite observations as ground truth, our GAN model achieves the synthesis of realistic precipitation maps at a resolution of $2.5^\circ \times 2.5^\circ$. While demonstrating proficiency in capturing prominent earth system patterns, the generated maps also reveal areas for improvement, underscoring the need for model refinement. These synthetic precipitation maps hold immense promise for diverse applications, including ensemble-based forecasting, missing data imputation, climate scenario studies, and understanding hydrometeorological responses to varied precipitation patterns.

Keywords Precipitation maps · Generative Adversarial Networks · Global Climate

1 Introduction

Precipitation maps are crucial in hydrometeorological studies, playing a pivotal role in the modeling of hydrological systems and understanding the repercussions of diverse precipitation patterns [Kidd and Huffman, 2011, Moral, 2010, Ushio et al., 2003]. The efficient forecasting of weather, a critical aspect of this discipline, forms the cornerstone of techniques such as ensemble forecasting, which explores multiple precipitation scenarios for robust predictions [Leutbecher and Palmer, 2008, Demargne et al., 2014]. The accuracy of these forecasts is paramount for effective water resource planning and management, offering insights into the availability and distribution of water resources. Moreover, in the realm of flood prediction, the ability to model hydrological systems based on various precipitation scenarios is instrumental in anticipating and mitigating the impacts of imminent disasters [Mai et al., 2022, Koya et al., 2023].

Despite the availability of global precipitation datasets derived from satellite observations and reanalysis, limitations persist, especially in the context of observation-based maps that can serve as ground truth [Sun et al., 2018]. To embark on large ensemble forecasting, the generation of synthetic datasets becomes imperative, often requiring computationally intensive earth system models [Krakauer and Fekete, 2014]. Additionally, the heterogeneity in resolution across existing datasets necessitates the cumbersome processes of downscaling or upscaling [Raje and Mujumdar, 2011, Zhang et al., 2020, Tripathi et al., 2006]. Herein lies the opportune intersection with cutting-edge machine learning algorithms, offering the promise of producing realistic precipitation maps with significantly reduced computational demands.

Among these algorithms, Generative Adversarial Networks (GANs) [Goodfellow et al., 2020, 2014] emerge as the preeminent generative models in the current landscape, celebrated for their remarkable success, especially in image synthesis [Brock et al., 2018]. Comprising a generator and a discriminator, these networks engage in adversarial training, with the former striving to create synthetic images and the latter discerning between real and generated data. In the

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context of this research, we harness the power of GANs to generate realistic synthetic precipitation maps. The training dataset comprises satellite-observed global precipitation maps, and the successful synthesis of realistic maps holds the potential to revolutionize ensemble forecasting.

This study presents a comprehensive exploration into the application of GANs for the generation of synthetic precipitation maps. In the subsequent sections, we delve into the existing research landscape, detailing the methods and experiments undertaken, and presenting the results obtained. By leveraging the prowess of GANs, we aim to address the complexities associated with precipitation mapping, paving the way for more efficient and resource-effective hydrometeorological studies.

2 Related work

In the evolving landscape of earth system modeling, the integration of novel machine learning algorithms has emerged as a transformative force [Nearing et al., 2021]. Recognizing the skill of these algorithms, several pioneering studies have sought to leverage their potential in generating precipitation maps. Early explorations predominantly embraced convolutional-based deep learning models, capitalizing on their inherent capacity to process image data [Kim et al., 2017, Shi et al., 2015, Ravuri et al., 2021, Luo et al., 2022]. Notably, Kim et al. [2017] and Shi et al. [2015] pioneered the application of Convolutional Long Short-Term Memory (ConvLSTM) networks for precipitation nowcasting, a technique focused on predicting precipitation maps for the next two hours or less [Smith and Austin, 2000, Browning and Collier, 1989]. Building on this foundation, Luo et al. [2022] further expanded the horizon by employing GANs with a convolutional LSTM generator for the same task.

In a more recent stride, Ravuri et al. [2021] proposed an innovative framework featuring a conditional GAN, equipped with a singular generator and dual discriminators—one for spatial consistency and another for temporal consistency. Their work encompassed the exploration of various deep learning architectures, including U-Net, with a comprehensive performance comparison. However, it is noteworthy that all these commendable endeavors were confined to producing precipitation maps for specific regions, overlooking the broader perspective essential for complete earth system modeling using ensembles.

To address this critical gap, our work takes a pioneering step by exploring the potential of GANs on a global scale. As far as our investigation reveals, there is a noticeable lack of previous efforts applying GANs to generate global precipitation maps. The significance of our contribution lies in extending the application of GANs to a global context, aiming to produce precipitation maps at a decent resolution—thus opening new perspectives for comprehensive earth system modeling and ensemble-based hydrometeorological studies.

3 Methodology

In the pursuit of advancing the field of precipitation mapping, this research article introduces a pioneering approach utilizing GANs to create synthetic precipitation maps of unparalleled realism. The GAN architecture, featuring a dynamic interplay between a generator and a discriminator network, undergoes rigorous adversarial training to refine its capability in generating precipitation patterns that faithfully replicate authentic meteorological phenomena.

3.1 Generator Network

The intricate operations of the generator network, depicted in Figure 1, commence with a latent noise vector of dimension 128, denoted as the latent dimension. This vector undergoes a transformative journey, passing through a linear layer, reshaping, and four sequential 2D deconvolutional layers [Zeiler et al., 2010]. The final output, carefully crafted to mirror the shape of original precipitation maps. At each stage of this journey, strategic application of batch normalization [Ioffe and Szegedy, 2015] techniques harmonizes the training dynamics, while rectified linear unit activation [Nair and Hinton, 2010] introduces the crucial non-linear element. The final output comes through a sigmoid activation, producing a value between zero and one for each grid (or pixel in the language of computer vision).

3.2 Discriminator Network

In contrast, the discriminator network, showcased in Figure 2, assumes the role of a discerning critic, trained at distinguishing between real precipitation maps and their synthetic counterparts created by the generator. Four 2D convolutional layers [Zeiler and Fergus, 2014], seamlessly integrated with leaky rectified linear units [Xu et al., 2015] exhibiting a negative slope of 0.2, form the backbone of the discriminator. Scrutinizing input images rigorously, the discriminator’s final output traverses from the convolutional layer through flattening to a linear layer. The resultant

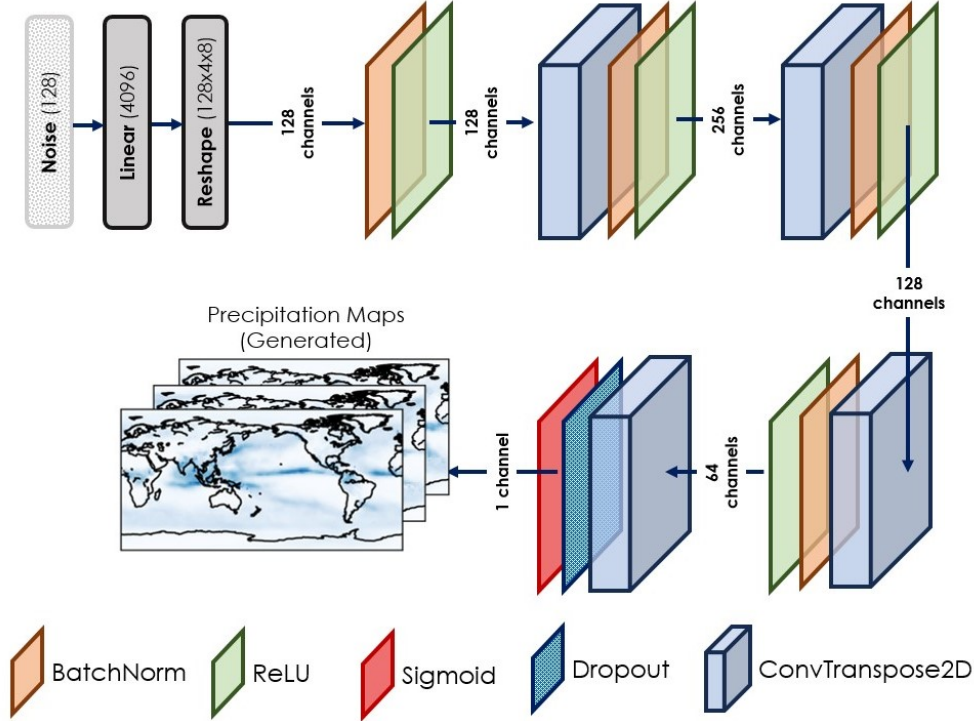


Figure 1: Generator network, consisting batch normalization, ReLU activation, dropout and 2D deconvolutional layers.

Kernel size, stride, and padding in first three deconvolutional layers are 4, 2, and 1 respectively. In the final deconvolutional layer, to match the output size to that of real precipitation maps, kernel size is set to 18, stride is set to 2, and padding is set to (4,0).

singular value provides the evaluation (i.e., a value between 0 and 1 representing fake to real) as it passes through a sigmoid activation. This value stands as the discerning judgment of the discriminator, providing a definitive assessment of the originality of the input image—whether it emerged from empirical meteorological data or from the generator’s synthesis.

The synergistic interplay between the generator and discriminator, orchestrated through adversarial training, propels the GAN toward a high proficiency in producing synthetic precipitation maps. Figures 1 and 2, meticulously crafted blueprints of the architectures within the generator and discriminator networks, vividly encapsulate the essence of this sophisticated methodology. This groundbreaking framework not only promises to redefine the landscape of precipitation mapping but also lays the foundation for future advancements in synthesizing complex geospatial phenomena through the lens of deep learning.

4 Experiment

In this pivotal experiment, we leverage precipitation maps sourced from the Global Precipitation Climatology Project (GPCP) as the ground truth to train and evaluate the GAN [Huffman et al., 1997, Adler et al., 2003]. GPCP, renowned for providing global estimates of mean precipitation at both daily and monthly frequencies, proves to be an invaluable dataset for our research. The monthly mean precipitation data spans an extensive period from 1979 to 2022 and is derived from a fusion of microwave imagers on polar-orbiting satellites and infrared imagers on geostationary satellites, integrated with rain-gauge observations. These maps, available at a resolution of $2.5^\circ \times 2.5^\circ$, translate into 72×144 grids (or pixels) for a single month’s map. Some of the sample maps are given in Figure 3. Note that the kernel size of the generator is adjusted to generate maps of this specific shape, ensuring seamless integration into our GAN framework.

The implementation of our GAN is realized through the PyTorch [Paszke et al., 2017] and Lightning [Falcon and The PyTorch Lightning team, 2019] libraries in Python [Van Rossum and Drake Jr, 1995], underscoring the contemporary and versatile toolset for deep learning. To prepare the precipitation maps for input into the network, a crucial normalization step between 0 and 1 is undertaken, aligning with the output range of the generator. The adversarial training process

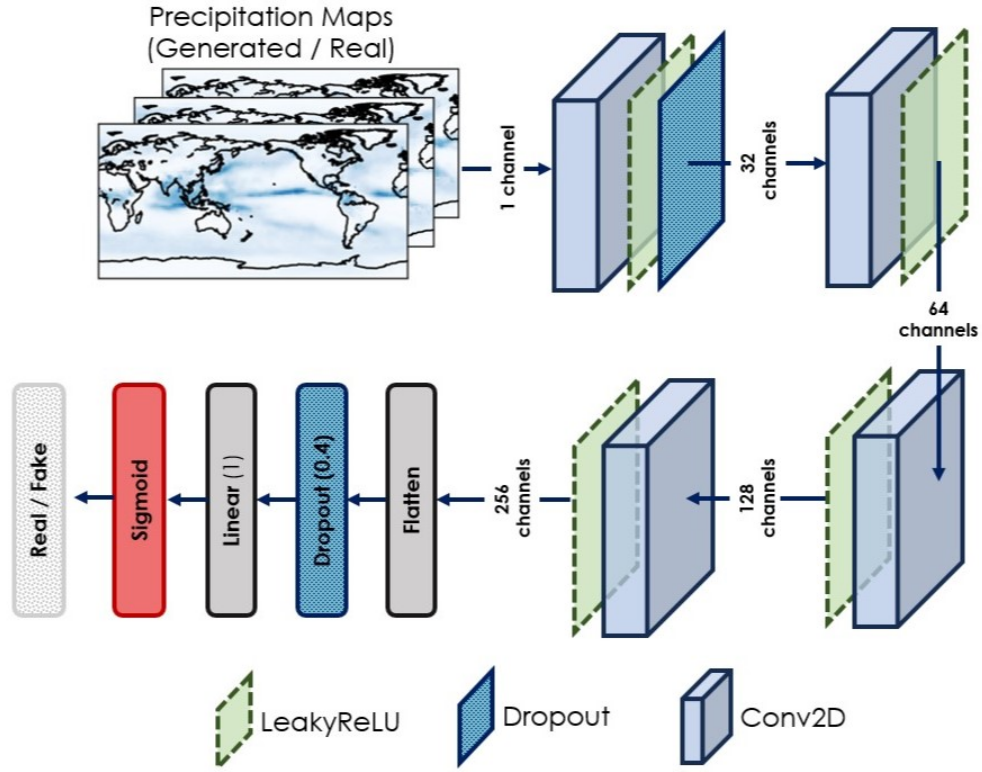


Figure 2: Discriminator network, consisting LeakyReLU activation, dropout and 2D convolutional layers. Kernel size, stride, and padding in all four convolutional layers are 5, 2, and 2 respectively.

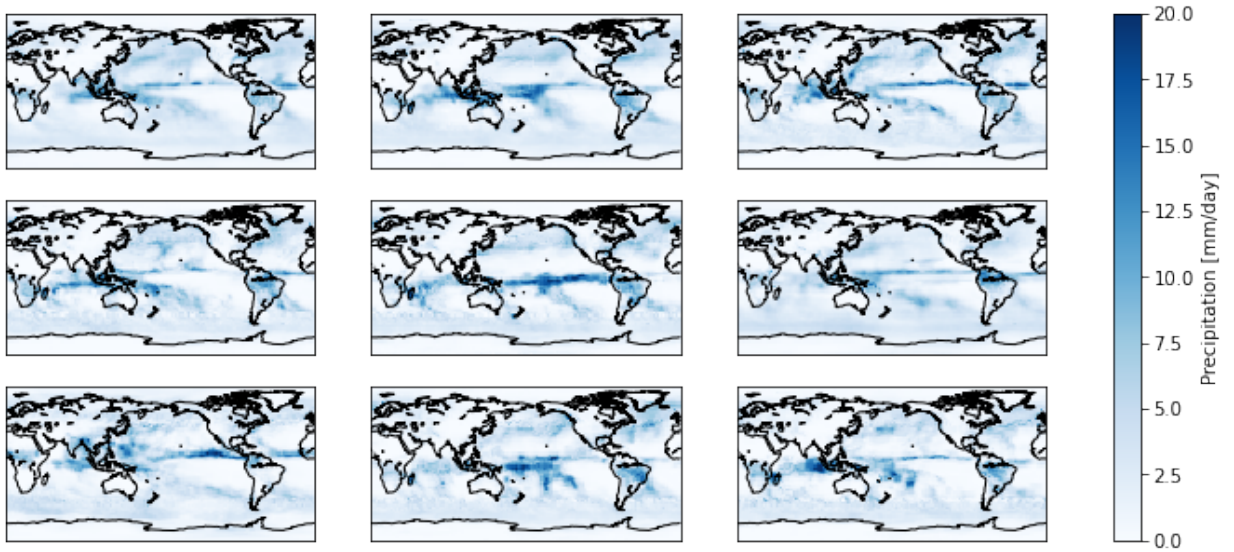


Figure 3: Observed precipitation maps from Global Precipitation Climatology Project.

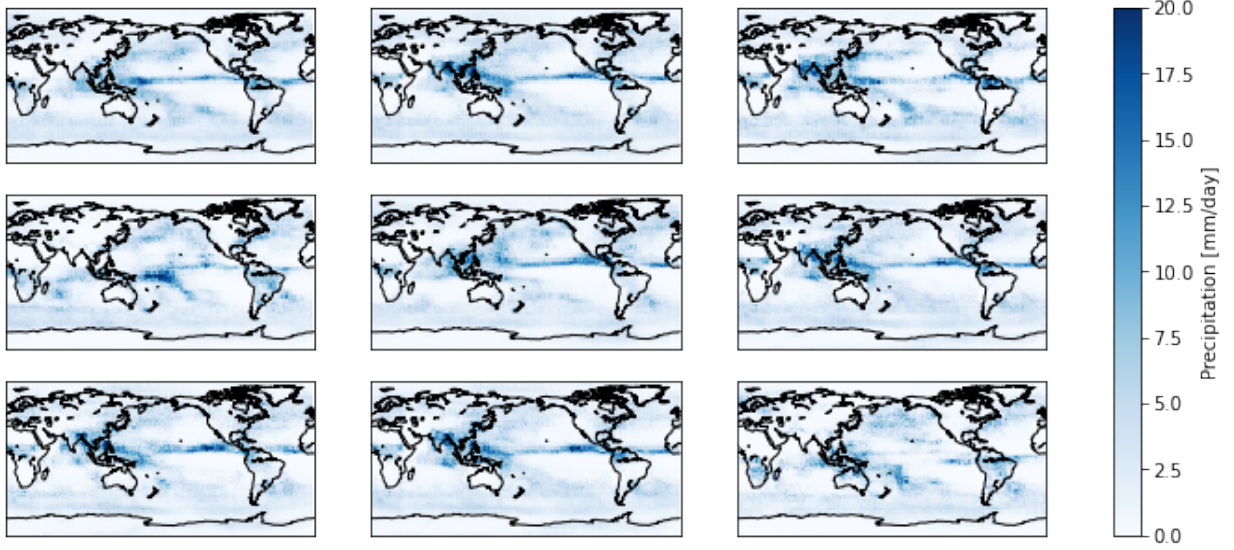


Figure 4: Synthetic precipitation maps generated by GANs.

involves the utilization of binary cross-entropy as the loss function, as depicted in Equation 1 and 2. In the generator loss (Equation 1), x represents real data samples, $G(x)$ represents generated data samples, and D is the discriminator. In the discriminator loss (Equation 2), x represents real data samples, z represents noise samples, $G(z)$ represents generated data samples, and D is the discriminator. The first term corresponds to the loss for real data, and the second term corresponds to the loss for generated (fake) data. Adam optimizers [Kingma and Ba, 2014] are employed for both the generator and discriminator networks, with a learning rate set at $2e-5$, fostering effective convergence during the training process. The GAN undergoes training for a duration of 100 epochs, with a modest batch size of 4, striking a balance between computational efficiency and model performance.

$$L_G = -\mathbb{E}_{x \sim p_{data}(x)} [\log D(G(x))] \quad (1)$$

$$L_D = -\frac{1}{2} \mathbb{E}_{x \sim p_{data}(x)} [\log D(x)] - \frac{1}{2} \mathbb{E}_{z \sim p_z(z)} [\log(1 - D(G(z)))] \quad (2)$$

Figure 4 showcases nine samples of the generated images, revealing an impressive likeness to real precipitation maps, as illustrated in Figure 3. The model skillfully captures intricate patterns over tropical regions, where substantial rainfall is observed. Particularly noteworthy is the quality with which the GAN reproduces the precipitation pattern in the tropical Pacific, connected to the prevailing El Niño-Southern Oscillation (ENSO) pattern. The model further excels in capturing precipitation patterns along the trade winds and westerlies, showcasing its ability to emulate complex meteorological phenomena. The utility of these synthetic maps extends across various domains. Beyond their application in augmenting existing precipitation datasets by creating additional synthetic examples, these maps can be invaluable for training machine learning models, especially in the realm of weather prediction. The need for large and diverse datasets for effective model generalization is met through this augmentation process. Moreover, the realistic nature of these synthetic maps positions them as effective tools for filling missing precipitation data, offering an innovative solution for enhancing the completeness of meteorological datasets. GANs empower us to generate an infinite number of synthetic maps, an attribute particularly useful for ensemble forecasting.

However, it is imperative to acknowledge the inherent imperfections in the generated maps. A close examination reveals that, in subtropical regions where precipitation is minimal throughout the year, the generated maps erroneously depict some precipitation pattern. This anomaly is notably evident over the Sahara Desert, where the introduction of generated rainfall contradicts the arid nature of the region. Consequently, reliance on these generated maps necessitates cautious scrutiny, particularly when applied to frameworks such as ensemble forecasting. Thorough validation and consideration of regional nuances are imperative before integrating synthetic precipitation maps into broader meteorological applications. The framework presented in this study lays a solid foundation, yet there exists a substantial scope for refinement and expansion to unlock its full potential. One promising avenue for improvement involves the incorporation of temporal information into the GAN architecture. Techniques such as positional encoding

could be seamlessly integrated, potentially enhancing not only the model’s performance but also enabling the production of precipitation maps specific to any desired time, thus adding a temporal dimension to the generated outputs.

Furthermore, a crucial enhancement lies in extending the framework to consider the influence of other earth system variables on precipitation patterns. Introducing dependencies on variables such as teleconnection indices (e.g., ENSO, NAO, PNA indices) into the map generation process holds the potential of capturing the interactions that drives precipitation dynamics. This expansion into multi-variable precipitation mapping could be aptly facilitated through conditional GANs, offering a structured and tailored approach to generate precipitation maps influenced by a spectrum of earth system factors. This evolution not only aligns with the intricacies of real-world meteorological phenomena but also broadens the utility of the generated precipitation maps in diverse applications, ranging from climate research to disaster preparedness. As we envision the future trajectory of this research, these enhancements will undoubtedly contribute to refining the precision and versatility of our framework, paving the way for a more comprehensive understanding of global precipitation dynamics.

5 Conclusion

In this research endeavor, we harnessed the cutting-edge capabilities of GANs, a pinnacle machine learning technique renowned for its ability in image generation. Our implementation incorporated a convolutional neural network-based generator within the GAN framework to produce synthetic precipitation maps on a global scale. The training leveraged monthly precipitation data derived from satellite observations, serving as a robust ground truth to refine the network’s capabilities.

Our results reveal the commendable achievement of generating realistic precipitation maps at a resolution of $2.5^\circ \times 2.5^\circ$, showcasing the GAN’s proficiency in capturing prominent patterns within the earth system. However, a discerning examination also unveiled the presence of some spurious patterns, underscoring the imperative for ongoing model refinement and enhancement.

The demonstrated performance of the GAN opens avenues for diverse applications of synthetic precipitation data. These synthesized maps, despite exhibiting areas for improvement, hold immense potential for tasks such as ensemble-based forecasting, imputing missing data, conducting climate scenario studies, and unraveling hydrometeorological responses to various precipitation patterns. As we navigate toward refining the model, this research not only contributes to the advancement of synthetic precipitation mapping but also lays the groundwork for a myriad of applications that can revolutionize how we perceive, analyze, and respond to global precipitation dynamics. The journey toward harnessing the full potential of GANs in this domain continues, promising further innovations and insights in the intersection of machine learning and hydrometeorology.

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